User Guide

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Based on our research paper, "Interpretable Landsat-to-Hyperspectral Dual Super-Resolution Without Large Matrix Inversion," we provide a demo file for researchers to investigate our theory and algorithm.

1 Requirements

The demo code of the proposed PAINT is conducted under Linux OS with NVIDIA RTX 3090 GPU, and the software environments are itemized as follows:

- Python: 3.10.13
- MATLAB: R2025b
- torch: 1.13.1+cu117
- scipy: 1.13.0
- torchvision: 0.14.1+cu117
- tensorboard: 2.20.0

2 Run the demo code

- Before running the demo, open demo.m and replace the "env_name" (line 19) with the name of your software environment.
- Run the script to obtain the PAINT-reconstructed hyperspectral image (HSI), and then evaluate its quantitative and qualitative performance.
- To evaluate PAINT on your own data, replace the ".mat" file in the "./main/data" folder with your own .mat data, where each .mat file must contain the AVIRIS HSI, Landsat multispectral image (MSI), and the Panchromatic image (Pan).

3 Model Training

- We also provide the training code in the "Train" folder to retrain PAINT on your own dataset.
- The motivation for releasing the training code is to enable users to train a country-specific model using your own regional datasets.

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- To train PAINT on your own dataset, place the dataset under the "./Train/dataset" directory. All data should be stored as .mat files containing paired MSI, HSI, and Pan images and converted to tensors. Then, organize the dataset into the corresponding training and validation subfolders (i.e., "./Train/dataset/Train_Spec" and "./Train/dataset/Valid_Spec").

- To start training, open a terminal and run "python Train.py". Model checkpoints are automatically saved during training.

4 Citation

If you find our work useful in your research or publication, please kindly cite our work:

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